Evolving the Geodetic Infrastructure to Meet New Scientific Needs (2020)

DETAILS

124 pages | 8.5 x 11 | PAPERBACK
ISBN 978-0-309-49778-7 | DOI 10.17226/25579

CONTRIBUTORS

Committee on Evolving the Geodetic Infrastructure to Meet New Scientific Needs; Committee on Seismology and Geodynamics; Board on Earth Sciences and Resources; Division on Earth and Life Studies; National Academies of Sciences, Engineering, and Medicine

SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine 2020. *Evolving the Geodetic Infrastructure to Meet New Scientific Needs*. Washington, DC: The National Academies Press. https://doi.org/10.17226/25579.

EVOLVING THE GEODETIC INFRASTRUCTURE TO MEET NEW SCIENTIFIC NEEDS

Committee on Evolving the Geodetic Infrastructure to Meet New Scientific Needs

Board on Earth Sciences and Resources

Committee on Seismology and Geodynamics

Division on Earth and Life Studies

A Consensus Study Report of

The National Academies of

SCIENCES · ENGINEERING · MEDICINE

THE NATIONAL ACADEMIES PRESS

Washington, DC

www.nap.edu

THE NATIONAL ACADEMIES PRESS • 500 Fifth Street, NW • Washington, DC 20001

This activity was supported by the National Aeronautics and Space Administration Grant No. 80NSSC18K0176 and the National Academy of Sciences' Arthur L. Day Fund. Any opinions, findings, conclusions, or recommendations expressed in this publication do not necessarily reflect the views of any organization or agency that provided support for the project.

International Standard Book Number-13: 978-0-309-49778-7
International Standard Book Number-10: 0-309-49778-7
Digital Object Identifier: https://doi.org/10.17226/25579

Additional copies of this publication are available from the National Academies Press, 500 Fifth Street, NW, Keck 360, Washington, DC 20001; (800) 624-6242 or (202) 334-3313; http://www.nap.edu.

Printed in the United States of America

Cover: Front: Illustration of the Icesat-2 satellite measuring sea ice thickness, an important climate change variable, in the Arctic. The sea ice height measurement depends on cm-accuracy laser range measurement as well as cm-accuracy tracking using the Global Navigation Satellite System (GNSS) and Satellite Laser Ranging (SLR) of the geodetic infrastructure. Back: The four geodetic measurement techniques of the geodetic infrastructure: Very Long Baseline Interferometry (top left), GNSS (top right), SLR (bottom left), and Doppler Orbitography and Radiopositioning Integrated by Satellite (bottom right). Images courtesy of NASA.

Suggested citation: National Academies of Sciences, Engineering, and Medicine. 2020. *Evolving the Geodetic Infrastructure to Meet New Scientific Needs*. Washington, DC: The National Academies Press. https://doi.org/10.17226/25579.

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

The **National Academy of Sciences** was established in 1863 by an Act of Congress, signed by President Lincoln, as a private, nongovernmental institution to advise the nation on issues related to science and technology. Members are elected by their peers for outstanding contributions to research. Dr. Marcia McNutt is president.

The **National Academy of Engineering** was established in 1964 under the charter of the National Academy of Sciences to bring the practices of engineering to advising the nation. Members are elected by their peers for extraordinary contributions to engineering. Dr. John L. Anderson is president.

The **National Academy of Medicine** (formerly the Institute of Medicine) was established in 1970 under the charter of the National Academy of Sciences to advise the nation on medical and health issues. Members are elected by their peers for distinguished contributions to medicine and health. Dr. Victor J. Dzau is president.

The three Academies work together as the **National Academies of Sciences, Engineering, and Medicine** to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The National Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.

Learn more about the National Academies of Sciences, Engineering, and Medicine at **www.nationalacademies.org**.

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

Consensus Study Reports published by the National Academies of Sciences, Engineering, and Medicine document the evidence-based consensus on the study's statement of task by an authoring committee of experts. Reports typically include findings, conclusions, and recommendations based on information gathered by the committee and the committee's deliberations. Each report has been subjected to a rigorous and independent peer-review process and it represents the position of the National Academies on the statement of task.

Proceedings published by the National Academies of Sciences, Engineering, and Medicine chronicle the presentations and discussions at a workshop, symposium, or other event convened by the National Academies. The statements and opinions contained in proceedings are those of the participants and are not endorsed by other participants, the planning committee, or the National Academies.

For information about other products and activities of the National Academies, please visit www.nationalacademies.org/about/whatwedo.

COMMITTEE ON EVOLVING THE GEODETIC INFRASTRUCTURE TO MEET NEW SCIENTIFIC NEEDS

DAVID T. SANDWELL (NAS), *Chair*, Scripps Institution of Oceanography, La Jolla, California
SRINIVAS BETTADPUR, The University of Texas at Austin
GEOFFREY BLEWITT, University of Nevada, Reno
JOHN J. BRAUN, University Corporation for Atmospheric Research, Boulder, Colorado
ANNY CAZENAVE, Centre National d'Etudes Spatiales, Toulouse, France
NANCY GLENN, Boise State University, Idaho, and University of New South Wales, Sydney, Australia
KRISTINE M. LARSON, University of Colorado Boulder (emeritus)
R. STEVEN NEREM, University of Colorado Boulder
MICHELLE SNEED, U.S. Geological Survey, Sacramento, CA
ISABELLA VELICOGNA, University of California, Irvine

National Academies of Sciences, Engineering, and Medicine Staff

ANNE LINN, Study Director, Board on Earth Sciences and Resources
ERIC EDKIN, Program Coordinator, Board on Earth Sciences and Resources

BOARD ON EARTH SCIENCES AND RESOURCES

ISABEL P. MONTAÑEZ, *Chair*, University of California, Davis
ESTELLA A. ATEKWANA, University of Delaware, Newark
BRENDA B. BOWEN, The University of Utah, Salt Lake City
CHRISTOPHER (SCOTT) CAMERON, Geological Consulting, LLC, Houston, Texas
NELIA W. DUNBAR, New Mexico Bureau of Geology & Mineral Resources, Socorro
RODNEY C. EWING (NAE), Stanford University, California
CAROL P. HARDEN, The University of Tennessee, Knoxville
ROBERT L. KLEINBERG (NAE), Institute for Sustainable Energy, Boston University, Massachusetts
THORNE LAY (NAS), University of California, Santa Cruz
ZELMA MAINE-JACKSON, Washington State Department of Ecology, Richland
MICHAEL MANGA (NAS), University of California, Berkeley
MARTIN W. McCANN, Stanford University, California
JEFFREY N. RUBIN
JAMES A. SLUTZ, National Petroleum Council, Washington, District of Columbia
SHAOWEN WANG, University of Illinois at Urbana-Champaign
ELIZABETH J. WILSON, Dartmouth College, Hanover, New Hampshire

National Academies of Sciences, Engineering, and Medicine Staff

ELIZABETH EIDE, Director
ANNE LINN, Scholar
DEBORAH GLICKSON, Senior Staff Officer
SAMMANTHA MAGSINO, Senior Staff Officer
NICHOLAS ROGERS, Financial Business Partner
COURTNEY DEVANE, Administrative Coordinator
ERIC EDKIN, Program Coordinator
RAYMOND (REMY) CHAPPETTA, Senior Program Assistant/Research Assistant

Acknowledgments

This Consensus Study Report was reviewed in draft form by individuals chosen for their diverse perspectives and technical expertise. The purpose of this independent review is to provide candid and critical comments that will assist the National Academies of Sciences, Engineering, and Medicine in making each published report as sound as possible and to ensure that it meets the institutional standards for quality, objectivity, evidence, and responsiveness to the charge. The review comments and draft manuscript remain confidential to protect the integrity of the process.

We thank the following individuals for their review of this report:

ZUHEIR ALTAMIMI, University of Paris
LAURA BOURGEAU-CHAVEZ, Michigan Tech Research Institute
DON CHAMBERS, University of South Florida
CLARA CHEW, University Corporation for Atmospheric Research
SHIN-CHAN HAN, University of Newcastle
SEAN HEALY, European Centre for Medium-Range Weather Forecasts
TOM HERRING, Massachusetts Institute of Technology
GEORGE HILLEY, Stanford University
MARTIN HORWATH, University of Dresden
PAUL SEGALL, Stanford University
YOLANDE SERRA, University of Washington
JOHN VIDALE, University of Southern California

Although the reviewers listed above provided many constructive comments and suggestions, they were not asked to endorse the content of the report nor did they see the final draft before its release. The review of this report was overseen by DANNY REIBLE, Texas Tech University, and KEITH CLARKE, University of California, Santa Barbara. They were responsible for making certain that an independent examination of this report was carried out in accordance with the standards of the National Academies and that all review comments were carefully considered. Responsibility for the final content rests entirely with the authoring committee and the National Academies.

Contents

SUMMARY 1

1 INTRODUCTION 9
Committee's Tasks and Approach, 9
Geodetic Infrastructure and Terrestrial Reference Frame, 12
Organization of This Report, 16
References, 17

2 PROGRESS IN MAINTAINING AND IMPROVING THE GEODETIC INFRASTRUCTURE 19
Maintaining and Improving the Geodetic Infrastructure, 19
Enhancing Specific SLR and VLBI Sites, 20
International Geodetic Network, 21
GNSS/GPS National Network, 21
International Geodetic Services and the ITRF, 22
Federal Geodetic Service, 22
Geodesy Workforce, 23
Summary, 24
References, 24

3 SEA-LEVEL CHANGE 25
Science Overview, 25
Sea-Level Change, 29
Thermal Expansion—Ocean Heat Storage, 32
Ice Sheets and Glacier Mass Changes, 32
Land Water Hydrology, 34
Vertical Land Motion, 35
Summary, 35
References, 37

4 TERRESTRIAL WATER CYCLE 39
Science Overview, 39
Elastic Loading, 40
Aquifer-System Compaction (Land Subsidence), 42
Surface Water Monitoring by Satellite Altimetry, 46
Water Cycle Monitoring with Satellite Gravity, 46
Calibration/Validation and GNSS-IR, 47
Summary, 47
References, 48

5 GEOLOGICAL HAZARDS: EARTHQUAKES AND VOLCANOES 51
Science Overview, 51
Required Measurements and Links to the Terrestrial Reference Frame, 54
Summary, 56
References, 57

6 WEATHER AND CLIMATE 59
GNSS for Atmospheric Sensing, 59
Improvements in Weather Models, 60
Reducing Uncertainty in Climate Projections, 66
Summary, 66
References, 68

7 ECOSYSTEMS 71
Vegetation Dynamics, 71
Lateral Transport of Carbon, Nutrients, Soil, and Water, 74
Global Soil Moisture, 77
Permafrost and Changes in the Arctic, 77
Summary, 80
References, 80

8 PRIORITIES FOR MAINTAINING AND ENHANCING THE GEODETIC INFRASTRUCTURE 83
Accuracy and Stability of the TRF, 83
Accuracy and Stability of Satellite Orbits, 84
Accuracy of the Low-Degree Geopotential Harmonics, 84
Augmentation of the GNSS Station Network, 85
Supporting Software, Models, Data, and Expertise, 85
Summary, 86
Reference, 87

APPENDIXES
A Science and Applications Traceability Matrixes 89
B Speakers and Workshop Participants 107
C Biographical Sketches of Committee Members 109
D Acronyms and Abbreviations 113

Summary

Satellite remote sensing is the primary tool for measuring global changes in the land, ocean, biosphere, and atmosphere. Over the past three decades, active remote sensing technologies have enabled increasingly precise measurements of Earth processes, allowing new science questions to be asked and answered. As this measurement precision increases, so does the need for a precise geodetic infrastructure.

The connections between the geodetic infrastructure and science applications are illustrated in Figure S.1. The geodetic infrastructure (level 1) comprises four measurement techniques used to accurately determine the Earth's orientation in space, its gravitational field, the trajectories of satellites in orbit around the Earth, and the positions of reference points on the Earth. Data from these reference points are used to define the terrestrial reference frame (level 2), a set of coordinates and velocities of stable reference points on the surface of the Earth, which are used to define the locations of all other sites. Other geodetic products (e.g., orbit determination; level 3) are used to generate and interpret high-precision data from Earth-orbiting missions (level 4). These missions provide the connection between the terrestrial reference frame and the geophysical observables (level 5), which are needed to help answer science questions (level 6).

Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space (NASEM, 2018; referred to hereafter as the Decadal Survey) identified high priority questions and associated space observational requirements to support Earth system science and applications for 2017–2027. Many of the science questions in the Decadal Survey can be supported by the existing geodetic infrastructure, as long as it is maintained. However, other science questions require enhancements to the infrastructure. For example, active remote sensing systems at the core of the National Aeronautics and Space Administration (NASA) program—such as Jason-3, NASA-Indian Space Research Organisation Synthetic Aperture Radar, Ice, Cloud, and land Elevation Satellite 2, Gravity Recovery Climate Experiment Follow On (GRACE-FO), and Surface Water Ocean Topography (SWOT)—often require more accurate timing and orbit information to achieve their threshold science requirements. Understanding and implementing improvements to the geodetic infrastructure and terrestrial reference frame is urgent because high-precision data needed for Decadal Survey science questions are already flowing from satellites in orbit.

At the request of NASA managers, the National Academies of Sciences, Engineering, and Medicine established a committee to summarize progress in maintaining and improving the geodetic infrastructure and to identify improvements to the geodetic infrastructure to meet new science needs laid out in the Decadal Survey. The committee tasks are given in Box S.1 and the responses to these tasks are summarized below.

TASK 1: PROGRESS IN MAINTAINING AND IMPROVING THE GEODETIC INFRASTRUCTURE

The committee's first task was to summarize progress and future aspirations for maintaining and improving the geodetic infrastructure, as detailed

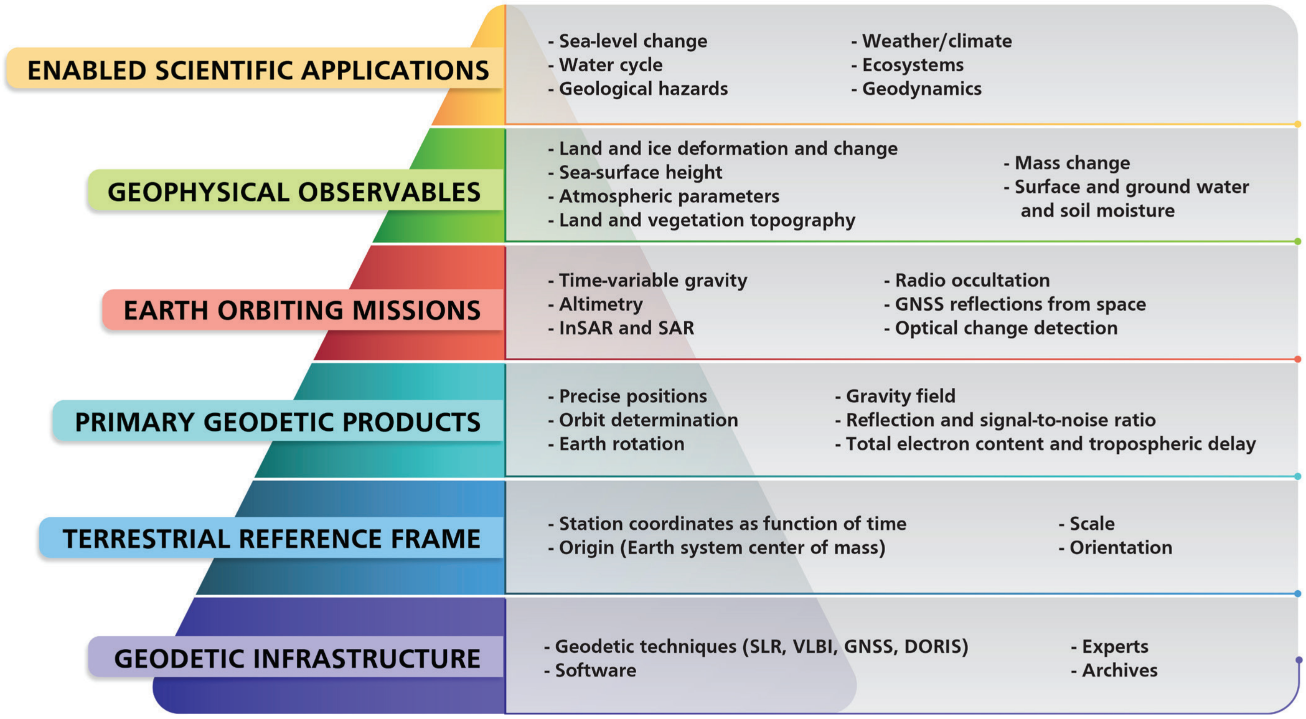

FIGURE S.1 Illustration of how the geodetic infrastructure is connected to enabled scientific applications. NOTE: DORIS = Doppler Orbitography and Radiopositioning Integrated by Satellite; GNSS = Global Navigation Satellite System; InSAR = Interferometric Synthetic Aperture Radar; SLR = Satellite Laser Ranging; VLBI = Very Long Baseline Interferometry.

in the recommendations in *Precise Geodetic Infrastructure: National Requirements for a Shared Resource* (NRC, 2010). The geodetic infrastructure includes the measurement systems and facilities that allow continuous collection of data at the reference points that define the terrestrial reference frame, as well as international geodetic services that play a role in the measurement systems or produce enabling data sets or models. Four complementary measurement techniques are used to define the reference frame parameters (origin, orientation, and scale), with each technique bringing specific strength to the reference frame definition:

1. Very Long Baseline Interferometry (VLBI), which provides information on Earth orientation angles and scale.
2. Satellite Laser Ranging (SLR), which provides information on the location of the center of mass of the Earth and scale. SLR is also a passive backup tracking method that can be used for orbit determination when other instruments (e.g., GNSS) fail.

BOX S.1
Committee's Tasks

1. Summarize progress in maintaining and improving the geodetic infrastructure, as detailed in the recommendations in *Precise Geodetic Infrastructure: National Requirements for a Shared Resource* (NRC, 2010), and aspirations for future improvements through, for example, new technology and analysis.
2. Identify science questions from the 2018 Decadal Survey on Earth Science and Applications from Space (NASEM, 2018) that depend on geodesy, and describe the connections between these questions, associated measurement requirements, and geodetic data.
3. Discuss the elements of these science questions that drive future requirements for the terrestrial reference frame, Earth orientation parameters, and satellite orbits, and identify what geodetic infrastructure changes are needed to help answer the questions.
4. Identify priority improvements to the geodetic infrastructure that would facilitate advances across the science questions identified in Task 2.

3. A network of Global Navigation Satellite System (GNSS) stations, installed much more densely over the globe than the small number of VLBI and SLR sites. The density of this network allows tens of thousands of GNSS receivers on spacecraft, aircraft, ships, and buoys, and in local geodetic arrays to access or connect to the International Terrestrial Reference Frame (ITRF). The GNSS network also makes a vital contribution to the measurement of polar motion.
4. Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS), which is mainly used to compute accurate orbits of altimetric spacecraft and to enhance the global distribution of ITRF positions and velocities.

A large number of U.S. federal agencies contribute to the development and maintenance of the geodetic infrastructure. NASA operates a set of VLBI and SLR sites and hosts a few DORIS sites. The U.S. Naval Observatory supports the operation and upgrade of U.S. VLBI stations and provides Earth orientation parameters that describe irregularities in the rotation of the Earth. NASA, the National Science Foundation (NSF), the National Oceanic and Atmospheric Administration's National Geodetic Survey, and the U.S. Geological Survey operate about one-quarter of the GNSS sites that form the core of the International GNSS Service. The National Geospatial-Intelligence Agency maintains a Global Positioning System (GPS) tracking network. In addition, U.S. federal agencies make substantial contributions to the international geodetic infrastructure through participation and leadership in international geodetic services.

The committee asked the above U.S. agencies to present their progress in and aspirations for maintaining and enhancing the geodetic infrastructure. Since the NRC (2010) report *Precise Geodetic Infrastructure: National Requirements for a Shared Resource* was published, several agencies have upgraded their networks (e.g., by replacing datums and upgrading GNSS sites to allow real-time streaming). Progress has been slower in modernizing VLBI and SLR systems. The committee found three areas of concern. First, the precision of the next-generation VLBI and SLR systems has not been validated with long-term data-driven studies (as opposed to simulation) in the refereed literature. Second, few VLBI or SLR stations have been added to complement and increase the density of the international geodetic network, especially in the southern hemisphere, leading to greater errors in the north-south location of the center of mass of the Earth. Third, a unified, highly accurate, national GNSS observing system has not been developed that could both (a) serve as the U.S. realization of and connection to the ITRF and (b) support the Decadal Survey science questions. Most of the geodetic networks operated by U.S. agencies have upgraded their GPS systems to receive signals from multiple satellite systems (multi-GNSS) or have clear plans to do so. However, plans to support the software and associated products (orbits and clocks) and models (e.g., location of antenna phase centers) needed for multi-GNSS data streams are not clear.

A broader concern is that, with an aging workforce and declining number of graduates trained in geodetic techniques and models, the United States risks losing its leadership role in geodesy or even its ability to meet the needs of U.S. geodesy programs. It is also at risk of losing redundancy (and hence validation capability) in the highest-grade geodetic data analysis software, independently written and maintained by more than one research group.

TASKS 2 AND 3: DECADAL SURVEY SCIENCE QUESTIONS THAT DEPEND ON THE GEODETIC INFRASTRUCTURE

Task 2 was to identify science questions in the Decadal Survey that depend on geodesy and to describe the connections between these questions, associated measurement requirements, and geodetic data. The committee selected a range of science questions that depend primarily on maintaining the current geodetic infrastructure (weather and climate and ecosystems) or on improving its capabilities (sea-level change, terrestrial water cycle, and geological hazards). Those science questions were discussed at a 2-day workshop in February 2019 attended by geodesists working to maintain and improve the geodetic infrastructure and discipline scientists seeking to answer questions that require an accurate terrestrial reference frame. Together, they identified what specific aspects of the geodetic infrastructure need to be maintained or improved to help answer the science questions being considered (Task 3). The science questions and their geodetic needs are summarized below.

Sea-Level Change

Sea level is a leading indicator of climate change because its long-term change is driven mainly by the amount of heat being absorbed by the oceans and the amount of land ice being melted by a warmer atmosphere and oceans. Monitoring sea-level changes at global to regional scales, understanding the causes of these changes, and projecting how sea level might change in the future are critical for mitigating adverse impacts on coastal infrastructure, ecosystems, and human society. A precise geodetic infrastructure is essential for studies of (1) absolute sea-level change (sea level measured with respect to the Earth's center of mass or other suitable reference surface), which is important for understanding climate change; and (2) relative sea level (sea level measured with respect to the possibly moving land surface), which is important for assessing the impacts along the coasts.

All of the measurements of sea-level change and its components (ocean thermal expansion, ice sheet and glacier mass change, land water hydrology, vertical land motion, and the effects of melting ancient and modern land ice) require a terrestrial reference frame that is accurately defined as a function of time. The terrestrial reference frame needs to have an accurately defined origin and be free of drifts and other errors, lest they create errors in the satellite measurements that could be misinterpreted as climate signals. This will become particularly challenging as the Earth's shape and gravity field change due to climate change. Of particular concern is the movement of the Earth's center of mass relative to the reference frame origin as the ice sheets melt, which could amount to several centimeters over the course of a century. In addition, geodetic sites near areas of ice mass loss may show anomalous motion and should be treated carefully if used to define the reference frame. It is also important to be able to reconstruct the terrestrial reference frame back in time, so that sea level measurements made a century from now can be compared to sea-level measurements made today or 25 years ago.

Terrestrial Water Cycle

Observing and understanding the water cycle and changes in the water cycle are essential for protecting this life-enabling resource both now and in the future. High-precision geodesy has become an important tool for hydrologists, climate scientists, and water managers, enabling a range of studies, including (1) elastic loading caused by changes in terrestrial water storage; (2) aquifer-system compaction and land subsidence caused by groundwater overdraft; (3) surface-water monitoring to support science, water management, and flood forecasting; and (4) water-cycle monitoring to track changes in total water storage and measure water cycle components (soil moisture, snow water equivalent, and vegetation water content).

The main geodetic focus of terrestrial water cycle applications is the ability to monitor absolute vertical deformation at local, regional, and continental scales. In the United States, this monitoring ability requires a backbone of core GNSS sites having a spacing of ~40 km and weekly Interferometry Synthetic Aperture Radar (InSAR) and altimetry acquisitions. Swath altimetry (e.g., SWOT) is needed to frequently measure surface water level (lakes and rivers), and is calibrated using tide gauges tied to the terrestrial reference frame by GNSS. The orbits of the InSAR and altimetry satellites rely on well-distributed GNSS stations at the surface of the Earth, as well as a stable and accurate terrestrial reference frame. Monitoring the water mass changes in the larger basins requires monthly time-variable gravity measurements from GRACE-type missions with support from the SLR network. Timely production and distribution of water cycle products relies on open data, accurate/open software, and a skilled workforce.

Geological Hazards

Earthquakes and volcanic eruptions open a window on processes operating within the Earth. They are also capable of great destruction, which has led to substantial efforts to forecast their occurrence and mitigate their impacts (e.g., reinforcing buildings to withstand expected shaking). Because earthquake and volcanic cycles occur on hundred- to thousand-year time scales, global and long-duration observations are needed to capture enough partial cycles to understand and model the underlying physical processes and so advance forecasting. The required measurements include surface deformation, time-variable gravity, surface topography, sea surface tsunami waves, and surface cover and atmospheric changes. All of these measurements depend on maintenance and moderate improvements of the geodetic infrastructure.

The surface deformation measurements depend on a global backbone of GNSS sites that is augmented with higher spatial resolution, but less frequent (weekly) InSAR measurements. The combined system should be able to monitor global plate motions at mm/yr accuracy with local strain rate measurements at sub 50 nanostrain/yr precision, which requires a slight enhancement in the GNSS network. Approximately 40 km or better spacing of geodetic-quality GNSS stations is needed for monitoring tectonically and volcanically active sites in North America. Accurate and near-real-time satellite orbits and clocks are needed for both long-term monitoring and disaster mitigation. A time-dependent terrestrial reference frame combined with time-dependent gravity will be needed to track deformations from major tectonic events, especially in ocean areas not monitored by GNSS and InSAR. Ocean GNSS sites, with real-time data delivery, can increase the accuracy of tsunami forecasts as well as provide platforms for seafloor geodesy. All of these applications rely on open data as well as accurate/open software and a skilled workforce to deliver reliable products in a timely manner.

Weather and Climate

The atmosphere is a complex system that varies spatially at length scales ranging from meters to the circumference of the Earth and time scales ranging from minutes and weeks (weather) to years and longer (climate). Understanding and predicting weather and climate requires high spatial and temporal sampling using a wide variety of sensitive terrestrial and space-based sensors combined with large numerical models that assimilate these data. Science applications that rely on maintenance or enhancement of the geodetic infrastructure include (1) improving weather models, and (2) monitoring climate and reducing uncertainty in climate projections.

These applications use ground-based GNSS to measure total column water vapor over land as well as space-based GNSS radio occultation to measure the vertical structure of the atmospheric water vapor and temperature over both land and ocean areas. The measurements rely on accurate clocks and orbits of the GNSS constellations, which in turn rely on the geodetic infrastructure. The sheer number of radio occultations per day requires a fully automated system with frequent updates of clocks and orbital information. Maintaining absolute accuracy over perhaps hundreds of years will require a stable terrestrial reference frame, accurate orbits for the GNSS satellites as well as the low-Earth orbiting satellites, and a consistent approach to antenna models and data processing.

Ecosystems

Ecosystems supply the services upon which all life depends. Understanding how ecosystems are changing and how these changes influence the Earth system are important for sustaining life on the Earth. Ecosystem science topics that use active remote sensing, and thus rely on the geodetic infrastructure, include (1) vegetation dynamics; (2) lateral transport of carbon, nutrients, soil, and water; (3) global soil moisture; and (4) permafrost and changes in the Arctic.

The main geodetic tools used to investigate ecosystems are (a) Synthetic Aperture Radar (SAR) and InSAR for estimating changes in vegetation land cover, lidar for measuring vertical biomass structure, bare-earth topography, and surface motion associated with erosional and depositional processes; and (b) GNSS-derived total column water vapor and radio occultation for measuring atmospheric water vapor and soil moisture. These tools rely on accurate satellite orbits and clocks and thus depend on maintaining the current accuracy of the geodetic infrastructure and terrestrial reference frame. The application of GNSS to ecosystem science is emerging, and so the signal-to-noise ratio from GNSS ground stations will need be archived to support future research. Sustained gravity measurements are also a priority. New geodetic needs include increasing the number of GNSS stations across environmental gradients and placing the stations at locations with tide gauges and soil moisture sensors. In addition, many more radio occultation measurements are needed to support water vapor observations.

TASK 4: IMPROVEMENTS TO THE GEODETIC INFRASTRUCTURE

Task 4 was to identify priority improvements to the geodetic infrastructure that would facilitate advances across the science questions summarized above. These improvements cover five main areas: (1) accuracy and stability of the terrestrial reference frame; (2) accuracy and stability of satellite orbits; (3) accuracy of the global-scale gravity field; (4) augmentation of the GNSS station network; and (5) analytical support for an enhanced geodetic infrastructure.

Most of the passive satellite systems recommended in the Decadal Survey rely on moderately accurate (<1 m) and near-real-time satellite orbits that are enabled by the continued maintenance of the geodetic infrastructure. In contrast, all of the active sensors that measure height (radar and laser altimetry), surface deformation (SAR), or path delay (radio occultation) require three-dimensional orbit accuracies that are better than or equal to the accuracy of the geophysical observable. For all of the satellite systems, active or passive, the availability of accurate orbits has enabled fully automated processing and accurate geolocation, which increases the exploitation of the large data sets being collected by Decadal Survey missions.

The accuracy and stability of satellite orbits relies on the accuracy and stability of the terrestrial reference frame, which is derived from the geodetic infrastructure. The committee identified three areas of improvement in the geodetic infrastructure needed to help answer the Decadal Survey science questions:

1. **Finalize deployment and testing of next-generation VLBI and SLR systems and complete deployment of multi-GNSS receivers to achieve a balance of geodetic measurement techniques between the northern and southern hemispheres, document the errors in the systems, and improve the ability to estimate their positions accurately and automatically.**
2. **Increase the capabilities for measuring the center of mass motions expected over the next 100 years, due to the melting of the Greenland and Antarctic ice sheets.**
3. **Work with the international community to implement a fully time-dependent terrestrial reference frame that will accommodate sudden, annual, and long-term changes in the locations of the fundamental stations.**

The most stringent requirements for enhancements to the accuracy and stability of the terrestrial reference frame are driven by science questions related to sea-level change, ice-mass loss, and land-surface deformation associated with (a) the movement of water over the surface of the land, cryosphere, and oceans; and (b) the elastic and viscoelastic response of the solid Earth to water loading, earthquakes, and volcanic eruptions.

Ground-based GNSS receivers are essential for achieving the Decadal Survey science objectives related to sea level, cryosphere, weather, climate, geological hazards, and ecosystems. The density of core GNSS stations in the United States needs to be increased in high priority regions, including plate boundary zones to capture the earthquake cycle, coastlines to capture land motion that could affect sea-level impacts and coastal ecosystems, and regions with substantial terrestrial water storage. In addition, the United States will need to work with the International GNSS Service to deploy additional GNSS sites in remote, rapidly deforming areas, such as the perimeters of the ice sheets that deform by changes in mass loading. Such sites need good stability of geodetic monuments, long duration, and high data rate and availability. The U.S. stations should be considered part of the U.S. geodetic infrastructure, open to everyone, and thus have long-term financial support. Many of these stations already exist, but the infrastructure is aging and users cannot rely on their continued operation by NSF.

Maintaining and enhancing the geodetic infrastructure to compute the terrestrial reference frame, satellite orbits, and other products requires complex software systems developed over decades by teams of scientists and engineers. The software systems ingest both the raw measurements from the geodetic infrastructure and models for the deformation of the Earth and for propagation of the electromagnetic waves through the ionosphere and atmosphere. Support for software is critical for using GNSS data to calibrate and validate future satellite missions. The most important aspects of this activity are that all of the raw data are completely open and that cross-checking occurs by at least two independent groups using largely independent and open software.

An important component of both the GNSS and InSAR infrastructure is the development of new software delivery tools to make these data seamlessly available to more users. The dramatic improvement in satellite orbits and clocks over the past decade has enabled automated processing of very large sets of repeated observations (e.g., SAR, optical, radar altimetry, and lidar) that was not possible just a few years ago. This advance is important because the data sets are too large for a human to be in the processing loop, and will require that the geodetic workforce work in close collaboration with the high performance computing community.

CONCLUDING REMARKS

The international geodetic infrastructure is the largely invisible foundation of Earth system science and applications. Most of the Decadal Survey science questions require maintenance of the geodetic infrastructure. However, key science questions—particularly those that need high-precision measurements from active remote sensing instruments—require enhancements to the geodetic infrastructure. Maintaining and in some cases enhancing the geodetic infrastructure will require collaboration among U.S. federal agencies and international partners as well as open data, accurate and open software, and a skilled geodetic workforce capable of developing and implementing improvements.

REFERENCES

NASEM (National Academies of Sciences, Engineering, and Medicine). 2018. Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space. Washington, DC: The National Academies Press.

NRC (National Research Council). 2010. Precise Geodetic Infrastructure: National Requirements for a Shared Resource. Washington, DC: The National Academies Press.

1

Introduction

Approximately every 10 years the National Aeronautics and Space Administration (NASA) asks earth scientists to reach a community consensus on a science and observations strategy for the next decade. *Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space* (NASEM, 2018) lays out high priority science questions and associated space observational requirements for atmosphere and climate, weather, hydrology, ecosystems, and solid earth science for 2017–2027. Underpinning these space observations and their interpretation is the geodetic infrastructure and its data products, notably the International Terrestrial Reference Frame (ITRF).

The connections between the geodetic infrastructure and science applications are illustrated in Figure 1.1 and Box 1.1. The geodetic infrastructure (level 1 of Figure 1.1) comprises four measurement techniques used to accurately determine positions of reference points on the Earth, the Earth's orientation in space, its gravitational field, and the trajectories of satellites in orbit around the Earth. Data from these reference points are used to define the terrestrial reference frame (level 2). Other geodetic data products (level 3) are needed to generate and interpret high-precision data from Earth-orbiting missions (level 4). These missions provide the connection between the terrestrial reference frame and the geophysical observables (level 5), which, in turn, are needed to answer science questions (level 6).

The existing geodetic infrastructure can support many of the science questions discussed in NASEM (2018), as long as it is maintained. However, enhancements to the geodetic infrastructure are required to support other Decadal Survey science questions, such as those connected with sea-level change. For example, Morel and Willis (2005) showed that changing the position of the center of mass of the Earth results in a commensurate change in sea level (see Figure 1.2). The Decadal Survey calls for the accuracy of regional sea-level rise to be better than 0.5 mm/yr decade, which requires a highly accurate and stable terrestrial reference frame. Enhancements to the geodetic infrastructure are also needed to analyze high-precision data from a variety of satellite sensors. Understanding what improvements to the geodetic infrastructure and terrestrial reference frame are needed is a matter of some urgency because high-precision data needed for Decadal Survey science questions are already flowing from satellites in orbit (e.g., Gravity Recovery and Climate Experiment Follow-On [GRACE-FO] and Ice, Cloud and land Elevation Satellite 2 [ICESat-2]). Improving the geodetic infrastructure would also facilitate new discoveries in earth sciences.

COMMITTEE'S TASKS AND APPROACH

At the request of NASA managers, the National Academies of Sciences, Engineering, and Medicine established a committee to identify key connections between geodesy and priority earth science questions, and to explore how to improve the geodetic infrastructure to meet new science needs. The committee tasks are given in Box 1.2.

This report builds on two previous National Academies reports. *Precise Geodetic Infrastructure: National Requirements for a Shared Resource* (NRC, 2010)

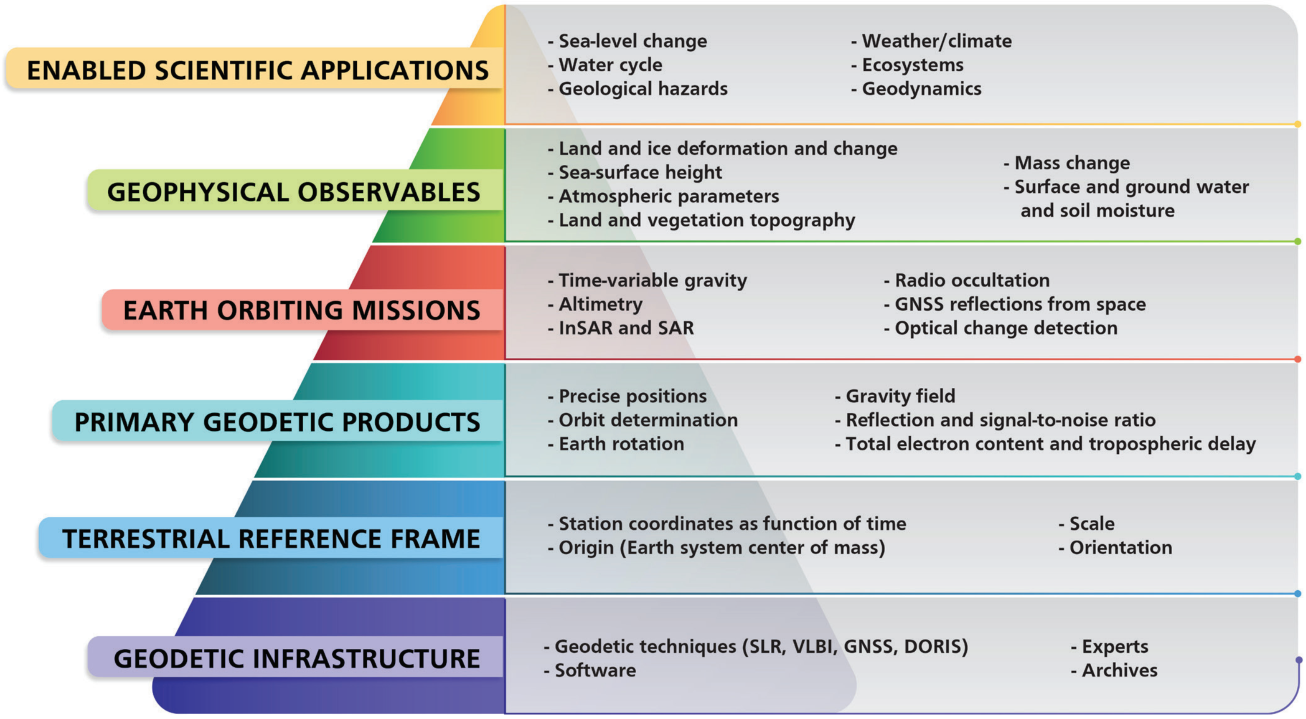

FIGURE 1.1 Illustration of how the geodetic infrastructure is connected to enabled scientific applications. NOTE: DORIS = Doppler Orbitography and Radiopositioning Integrated by Satellite; GNSS = Global Navigation Satellite System; InSAR = Interferometric Synthetic Aperture Radar; SLR = Satellite Laser Ranging; VLBI = Very Long Baseline Interferometry.

assessed the benefits of the geodetic infrastructure and recommended improvements to meet user demands for increasingly greater precision. To address Task 1, the committee invited U.S. federal agency managers responsible for the geodetic infrastructure to present their assessment of progress made in implementing the NRC (2010) recommendations and their aspirations for the future. The committee used material from the agency presentations, subsequent discussions with other experts, and its own expertise to address the task.

The second foundation report was *Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space* (NASEM, 2018), which listed the science questions to be considered in this study. To address Task 2, the committee combed through the science questions in NASEM (2018) and selected the ones that depend either on maintaining the current geodetic infrastructure or improving its capabilities.

Those science questions were discussed at a 2-day workshop in February 2019 that brought together those who maintain and improve the geodetic infrastructure with scientists from multiple disciplines seeking to answer questions that require an accurate terrestrial reference frame. The workshop had two goals. The first goal was to identify what specific aspects of the geodetic infrastructure need to be maintained or improved to help answer the science questions being considered (Task 3). Workshop participants considered future needs for ground networks, data processing, on orbit requirements, space-based approaches, and tools, such as simulation capabilities to quantitatively assess the impact of reference frame improvements. The second goal was mutual education: the scientists would better understand how their research connects with the underlying terrestrial reference frame, and NASA and other federal agencies would better understand how terrestrial reference frame realizations need to evolve to answer priority science questions.

The results from the first meeting (Task 1) and the workshop (Tasks 2 and 3) were used to identify priority improvements to the geodetic infrastructure that would facilitate advances across the science questions (Task 4).

BOX 1.1
The Satellite Orbit Connects the Geophysical Measurement to the Terrestrial Reference Frame

There are two basic types of satellite remote sensing instruments: passive and active. A passive sensor, like a camera, uses reflected sunlight to measure the intensity of each pixel of the image. An active sensor, such as an altimeter, sends a pulse of light toward the Earth. The pulse reflects from the land, ocean, or atmosphere and the sensor measures the two-way travel time (see Figure 1.2a). Using the speed of light, the travel time is converted to a range. Thus, active sensors measure both range and intensity, whereas passive sensors measure just the intensity.

The range measured by an active remote sensing satellite is only one half of the geophysical measurement (e.g., sea-surface height); the other half is the satellite orbit. Thus, orbit error maps directly into the error in the geophysical measurement. The orbit accuracy depends on the accuracy of the tracking system as well as the accuracy of the terrestrial reference frame. Because accurate orbits are essential for several Decadal Survey satellite missions, combined Global Navigation Satellite System (GNSS) and Satellite Laser Ranging (SLR) tracking are used to reduce the orbit error as well as to provide backup should one tracking instrument fail. A classic example of such a failure occurred in 1991, when the European Space Agency's ERS-1 satellite lost its primary PRARE tracking system. (Global Positioning System [GPS] tracking was not fully developed at the time.) The backup SLR system saved the day, and ERS-1 became one of the most important Interferometric Synthetic Aperture Radar and altimeter missions to date.[a]

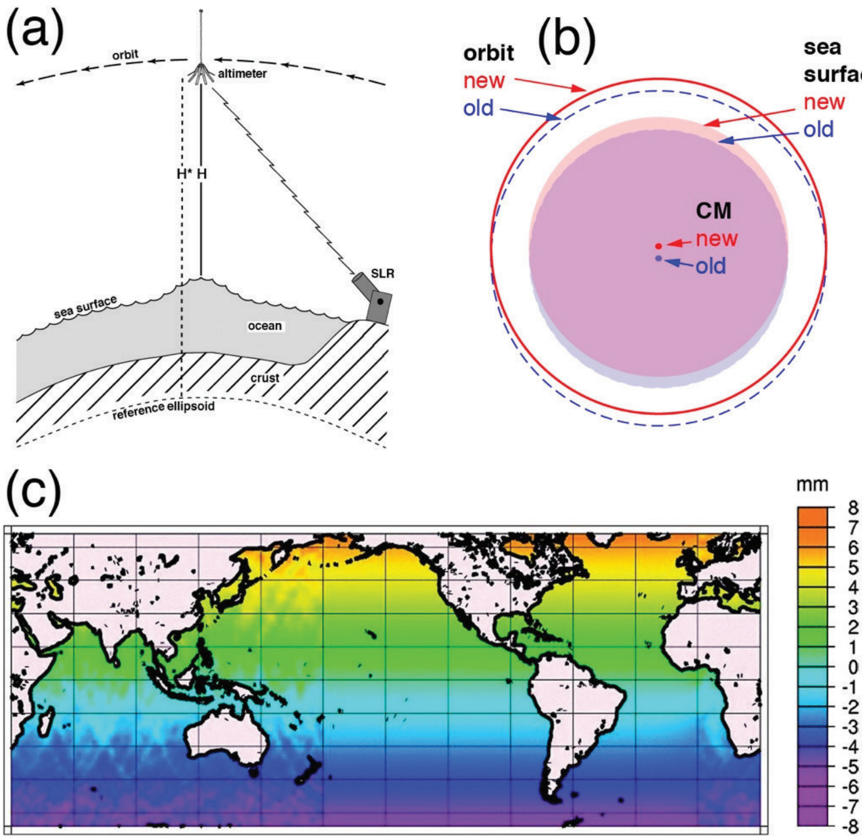

FIGURE 1.2 (a) The altimeter measures its height above the sea surface *H* by measuring the two-way travel time of a reflected radar pulse. The height of the satellite above the reference ellipsoid *H** is determined from the satellite orbit, which is measured by GNSS and SLR tracking. The sea-surface height is the difference between these two heights. SOURCE: Modified from Tapley et al., 1982. (b) The satellite orbits the center of mass (CM) of the Earth. A poleward shift in the CM causes a poleward shift in the orbit, which, in turn, results in a poleward shift in the sea-surface height. (c) Global change in sea-surface height caused by a 10 mm Z-translation of the center of mass of the terrestrial reference frame. While this is an extreme case based on the ITRF accuracy in 2005, the same direct connection between the terrestrial reference frame, orbit, and sea level holds today. SOURCE: Morel and Willis, 2005.

[a] See https://ilrs.cddis.eosdis.nasa.gov/missions/satellite_missions/past_missions/ers1_general.html.

BOX 1.2
Committee's Charge

1. Summarize progress in maintaining and improving the geodetic infrastructure, as detailed in the recommendations in *Precise Geodetic Infrastructure: National Requirements for a Shared Resource* (NRC, 2010), and aspirations for future improvements through, for example, new technology and analysis.
2. Identify science questions from the 2018 Decadal Survey on Earth Science and Applications from Space (NASEM, 2018) that depend on geodesy, and describe the connections between these questions, associated measurement requirements, and geodetic data.
3. Discuss the elements of these science questions that drive future requirements for the terrestrial reference frame, Earth-orientation parameters, and satellite orbits, and identify what geodetic infrastructure changes are needed to help answer the questions.
4. Identify priority improvements to the geodetic infrastructure that would facilitate advances across the science questions identified in Task 2.

GEODETIC INFRASTRUCTURE AND TERRESTRIAL REFERENCE FRAME

Terrestrial Reference Frame

A terrestrial reference system is a spatial reference system attached to the rotating Earth, and it includes the specification of its origin (usually at the center of mass of the Earth), its principal directions (connected with the equator or rotation axes and prime meridian), and a length scale. A terrestrial reference frame is the realization of the terrestrial reference system through a set of coordinates and velocities of stable reference points on the surface of the Earth whose positions are very accurately known as a function of time.[1] Such reference points are the locations of GNSS, SLR, Very Long Baseline Interferometry (VLBI), and Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS) tracking stations. Use of satellite tracking data from these locations results in satellite orbit positions that are expressed in that particular realization of the terrestrial reference frame.

Adoption and use of common terrestrial reference systems and frames allow diverse geodetic measurements to be linked over space and time (see Figure 1.3). The reference points used to realize the terrestrial reference frame are selected so that they have steady and predictable motions on the Earth's surface at time scales ranging from months to decades. Consequently, the frame itself evolves slowly and predictably and thus can be used for several years without a major update.

The quality of positioning within the terrestrial reference frame is described in terms of precision, accuracy, stability, and drift (NRC, 2010; see Box 1.3). The accuracy of the terrestrial reference frame can be specified—by quantifying uncertainty or by comparing two reference frame realizations—using seven parameters and their time variations. These parameters are the origin (three translations), the orientation (three rotation angles), and the scale (scalar). The science requirements on the accuracy or stability of several of these reference frame parameters drive the future geodetic infrastructure needs.

An international terrestrial reference system has been adopted by the International Earth Rotation and Reference System Service (IERS). The IERS, in collaboration with multi-technique services of the International Association for Geodesy, is also responsible for obtaining ITRF realizations. The ITRF realizations are updated as new data are added and as new technologies or new analysis methods are incorporated. The latest such realization is the ITRF2014 (Altamimi et al., 2016), and preparations have begun for the ITRF2020. Although other global reference systems exist (e.g., WGS84), the ITRF is regarded as having the greatest quality and is the most widely disseminated. The international earth science community, including NASA space-mission data providers and data users, have long used the ITRF for consistent earth science data analyses and interpretation. Consequently, in this report, the ITRF is used as the reference frame realization relevant to the Decadal Survey science objectives, and the phrase "terrestrial reference frame" refers to the ITRF.

[1] Presentation by Frank Lemoine, NASA, at the February 2019 workshop.

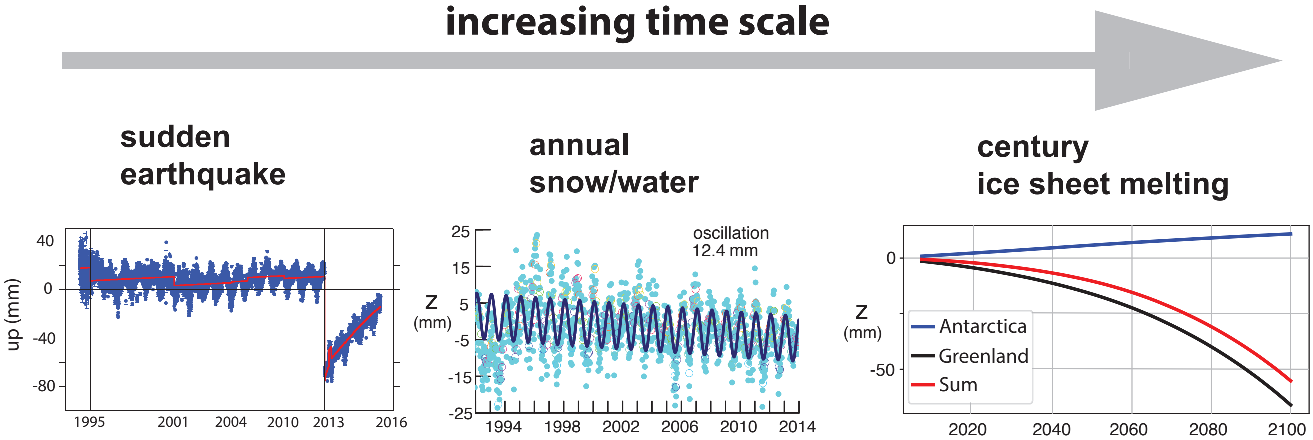

FIGURE 1.3 The Earth-fixed coordinates of the sites defining the terrestrial reference frame change on a wide range of time scales when mass is redistributed over the surface of the Earth. (Left) Sudden mass redistribution is caused by earthquakes and their postseismic response. SOURCE: Altamimi et al., 2016. (Middle) Annual variations in snow and water loading cause mainly Z-oscillations in the center of mass. SOURCES: Don Argus and colleagues, Jet Propulsion Laboratory, based on Altamimi et al., 2016. (Right) Melting of the ice sheets on century time scales causes nonlinear motions in the center of mass. The latter changes must be monitored for at least 100 years so sea level measured in 2100 can be compared with sea level measured in 2000. SOURCE: Modified from Adhikari et al., 2015.

BOX 1.3
Terms Used to Describe the Quality of Positioning Within a Reference Frame

Accuracy—how close a station position within a reference frame is to the truth. Precision contributes to accuracy, but accuracy also takes into account systematic biases arising from calibration errors or imperfect observation models.

Drift—the relative rotation, translation, and scale between different reference frames, which results in different velocities between stations given in each frame. Drift results from instability in one or both of the frames being compared, which in turn may result from systematic error in the measurement techniques, lack of precision in the measurements, or differences in the station motion models.

Precision—the ability to repeat the determination of position within a reference frame. Precision is necessary to resolve changes in position over time, and it is measured using statistical methods on samples of estimated positions. The precision of a reference frame itself refers to the variation in the reference frame parameters (origin, orientation, and scale) that arise from statistical variation in the data used to define the frame.

Stability—the predictability of the reference frame and the positions of the stations used to define the frame. In a stable reference frame, the defining parameters behave in a consistent manner, with no discontinuities over the time span of the geodetic observations. Furthermore, the ITRF should remain internally consistent, even as it is updated from time to time. Local site stability typically implies that all stations at that site do not move relative to each other, and that the site does not have nonlinear motions relative to the ITRF.

SOURCE: Abstracted from NRC, 2010.

Geodetic Infrastructure

The geodetic infrastructure includes the physical infrastructure (e.g., measurement systems and facilities) that allows continuous collection of data at the reference points that define the terrestrial reference frame, as well as geodetic services that play a role in the measurement systems or provide enabling data sets or models. Four complementary measurement techniques are used to define the time-dependent ITRF (see Figure 1.4), and their primary contributions include the following:

1. VLBI, which provides information on the three Earth orientation angles and scale.
2. SLR, which provides information on the location of the center of mass of the Earth and scale.
3. A network of GNSS stations, which enables densification of the reference frame, and provides supplementary information on all seven parameters of the terrestrial reference frame. The density of this network, compared with the relatively small number of VLBI and SLR sites, allows tens of thousands of GNSS receivers on spacecraft, aircraft, ships, and buoys and in local geodetic networks to access or connect to the ITRF (including in real-time). The GNSS network also makes a vital contribution to the measurement of polar motion.
4. DORIS, which is a ground-based beacon system mainly used for computing accurate orbits of altimetric spacecraft and for enhancing the global distribution of ITRF positions and velocities.

Each of these four measurement techniques makes several contributions to the terrestrial reference frame (see Table 1.1). These measurement techniques also underpin determination of satellite orbits and Earth orientation parameters.

Very Long Baseline Interferometry

The VLBI system comprises 47 radio telescopes that contribute to the measurements of the Earth's orientation and scale (Nothnagel et al., 2017; see

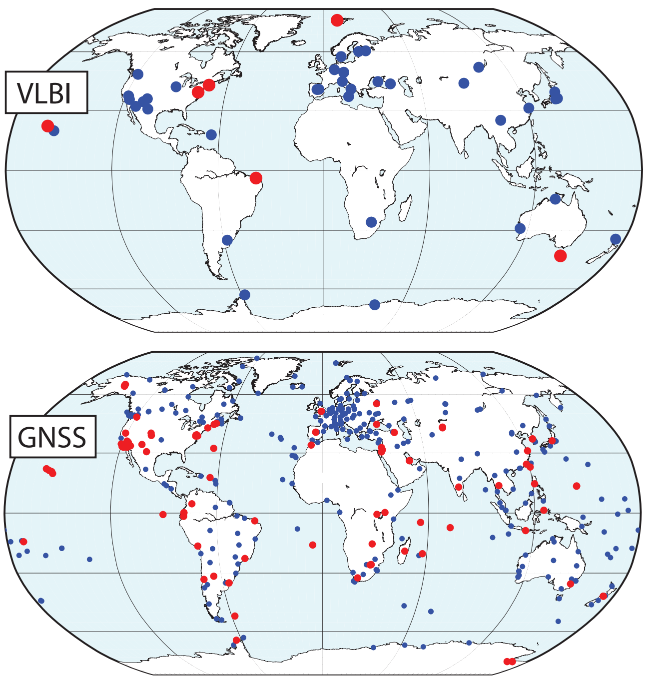

FIGURE 1.4 Positions of stations in the four measurement techniques that currently contribute to the ITRF. (Upper left) VLBI sites (47), including 6 sites (red) operated by NASA. (Upper right) SLR sites (39), including 8 sites (red) operated by NASA. (Lower left) GNSS sites (496) that form the core of the International GNSS Service. Of these, 122 sites (red) are operated by U.S. institutions, including NASA, the National Science Foundation, the National Geodetic Survey, the U.S. Geological Survey, and the National Geospatial-Intelligence Agency. (Lower right) DORIS sites (55), which are operated by the Centre National d'Etudes Spatiales. Three of these sites (red) are hosted by NASA. SOURCE: Data from Carey Noll, Secretary of the International Laser Ranging Service Central Bureau, 2019.

TABLE 1.1 Relative Contributions of Geodetic Measurement Techniques to the Terrestrial Reference Frame

Technique	VLBI	SLR	GPS	DORIS
Signal Target	Microwave quasars	Optical satellites	Microwave satellites	Microwave satellites
Observation type	Time difference	2-way range	Δ Range	Doppler
Celestial Frame (UT1)	**Strong**	Weak	Weak	Weak
Scale	**Strong**	**Strong**	Medium	Medium
Geocenter	Weak	**Strong**	Medium	Medium
Geographic Density	Weak	Weak	**Strong**	Medium

SOURCES: Don Argus, Jet Propulsion Laboratory, based on Altamimi et al., 2016, and Haines et al., 2015.

Figure 1.4). With this system, coordinated operation of two or more telescopes allows simultaneous recording of signals from the same extragalactic radio sources. The signal recordings are then pairwise cross-correlated between the telescopes to establish their changing positions with respect to the "fixed" celestial reference frame, as defined by the extragalactic radio sources. Ideally a large number of globally well-distributed VLBI stations should consistently participate in tracking sessions. Systematic errors in VLBI data and data products that can affect the stability of the terrestrial reference frame include the effects of gravitational deformation of VLBI antenna and tropospheric refraction errors.

VLBI is inherently a collaborative global activity, and the master schedule of observations is coordinated by the International VLBI Service for Geodesy and Astrometry (IVS; 41 institutions in 21 countries). This schedule is based on the availability of each station as well as the need for global and temporally dense sampling to measure daily changes in the rotation and orientation of the Earth. All VLBI observations are shared, and several centers of the IERS routinely produce Earth orientation parameters, such as the difference between Universal Time, which is defined by the Earth's rotation, and Coordinated Universal Time, which is defined by a network of precision atomic clocks. Predictions of this time difference are needed for many applications, such as satellite tracking and military operations.

NASA operates and maintains seven large radio telescopes at six sites around the Earth and is thus a major contributor to the ITRF. Moreover, NASA Goddard Space Flight Center currently hosts the IVS, which coordinates global operations up to 1 year in advance (Nothnagel et al., 2017). The NASA sites have been in operation since the 1980s and work is in progress to install the next-generation VLBI system (see Chapter 2).

Satellite Laser Ranging

The SLR system comprises 39 ground stations distributed around the Earth as well as 11 dedicated geodetic satellites (Pearlman et al., 2002, 2019; see Figure 1.4). The ground stations use short-pulse lasers, optical receivers, and accurate timing to measure the two-way travel time (and hence distance) to retroreflector arrays on the geodetic satellites. The geodetic satellites are mostly in high-altitude orbits where atmospheric drag and other nonconservative forces are minimal, ensuring a long lifetime in orbit. The SLR tracking data is sensitive both to the position of the center of mass of the Earth and to the large spatial scale variations in the gravity field. Such information is critical to the maintenance of the terrestrial reference frame. The biases in timing, range biases in tracking systems, and uncertainty in the knowledge of center of mass of the satellites carrying the retroreflectors can affect the quality of estimation of the reference frame parameters from SLR.

While several scientific satellite missions (e.g., ICESat-2, GRACE-FO, Jason-3, and NASA-ISRO Synthetic Aperture Radar) that support the Decadal Survey (NASEM, 2018) science questions normally use GNSS receivers for precise measurements of the orbital position, SLR provides independent validation of the centering and stability of the orbits for satellites orbits. SLR tracking also serves an important role as a backup tracking system in case of GNSS failure on Earth observation missions, and for determination of long-wavelength gravity field variations. As a result, the SLR ground stations routinely track more than 90 satellites, including much of the GNSS constellations, and thus provide an important link between the ITRF and satellite positions.

NASA currently operates 8 of the 39 global SLR stations. SLR operations, schedules, and products are coordinated by the International Laser Ranging Service (ILRS), which is currently located at NASA Goddard Space Flight Center. Weekly station coordinate solutions are developed at six ILRS analysis centers and combined as input to the ITRF.

Global Navigation Satellite System

The International GNSS Service (IGS) network comprises 496 globally distributed stations operated by a federation of more than 200 self-funded agencies, universities, and research institutions in more than 100 countries (see Figure 1.4). NASA's Jet Propulsion Laboratory operates 51 of the IGS stations and hosts the IGS Central Bureau. The IGS organizes the global GNSS network used to compute accurate GNSS orbits and clocks. Station coordinates from this network are an important contributor to the ITRF. The IGS orbits are available in real-time, rapidly (17-hour latency) and in post-analysis (13 days) time frames for GNSS orbits and Earth orientation parameters (polar motion). These frame products are used by continuously operating GNSS receivers (currently more than 10,000 receivers of geodetic quality) around the world, as well as by surveyors, aircraft, and NASA satellites. The IGS also provides other products, such as troposphere delays and maps of the variations in the Earth's ionosphere. All IGS products are provided without restriction.

In the committee's view, the GNSS infrastructure is not limited to the IGS stations, but also includes GNSS stations that have long duration and stability and are needed to meet the science objectives of this report.

Doppler Orbitography and Radiopositioning Integrated by Satellite

The DORIS system comprises approximately 55 autonomous and globally distributed stations that have been managed and deployed by the Centre National d'Etudes Spatiales and the Institut Géographique National since 1986 (Moreaux et al., 2016; see Figure 1.4). The third generation of antennae ("Starec C") is now being deployed. DORIS receivers are used primarily on altimeter satellites (Topography Experiment, Jason 1–3, Environmental Satellite, Cryosat-2, Sentinel-3A/B, and HY-2A) to provide real-time positions with ~30 mm radial orbit accuracy. Co-location of DORIS beacons with other satellite tracking techniques and cohosting other tracking instruments with DORIS onboard these altimetric satellites allows the altimetric sea-level measurements to be interpreted in the ITRF with confidence. Systematic errors in the solar radiation pressure modeling on spacecraft can affect the estimation of parameters of the ITRF.

The International DORIS Service (IDS) provides data and products to geodetic, geophysical, and other research and operational groups. Seven analysis centers contribute their time-dependent station positions and tracking data for the development of the ITRF.

Geodetic Services

Generation of the ITRF starts by distributing the raw data from the geodetic measurement systems discussed above to the analysis and combination centers (IVS, ILRS, IGS, and IDS), where it is analyzed and refined using computer models and statistical analyses (see Figure 1.5). These higher-level products are then distributed to the IERS to develop the ITRF. No single country or agency is responsible for generating these products. Instead, all parties involved work in an open international collaborative environment to provide the most accurate reference frame for science and applications. Several U.S. agencies, described in Chapter 2, contribute to and benefit from this global activity.

The geodetic services also play an important role in meeting the Decadal Survey (NASEM, 2018) science questions. All NASA missions that rely on accurate orbits for data collection and interpretation depend on the services providing GNSS satellite ephemerides and Earth orientation parameters (at various latencies) as key enabling or ancillary data sets. The services also test, establish, and disseminate the data processing models and standards to the community, which promotes harmonization across diverse space missions.

ORGANIZATION OF THIS REPORT

This report discusses the geodetic infrastructure needed to meet new science needs. Chapter 2

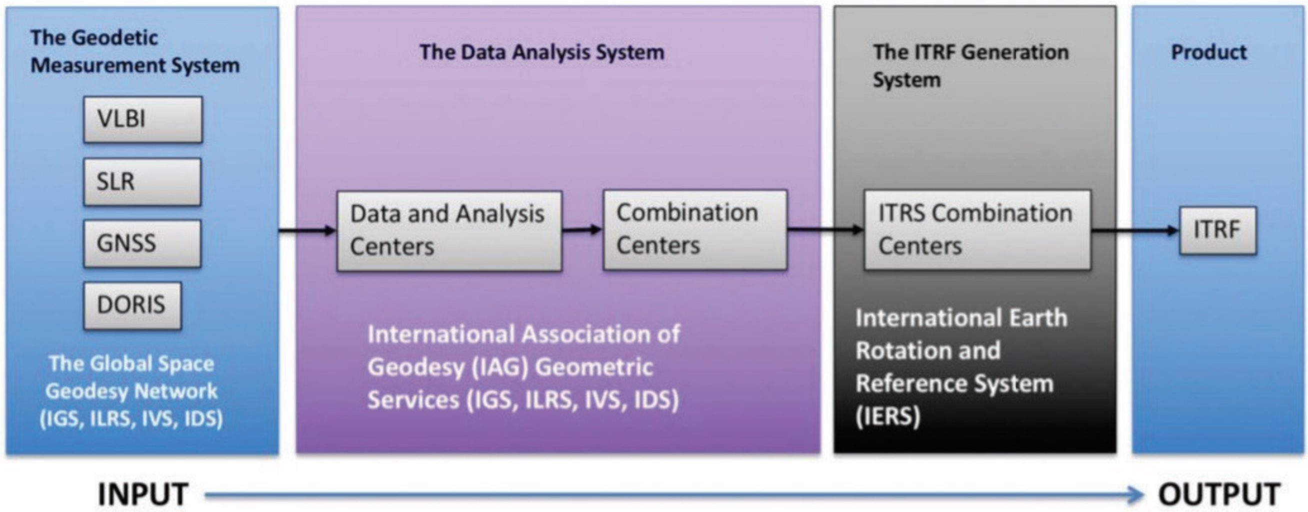

FIGURE 1.5 Schematic diagram of the production of the ITRF using geodetic measurements from VLBI, SLR, GNSS, and DORIS. All of the associated geodetic services are organized under the International Association of Geodesy. NOTE: IDS = International DORIS Service; IGS = International GNSS Service; ILRS = International Laser Ranging Service; IVS = International VLBI Service for Geodesy and Astrometry. SOURCE: Frank Lemoine, NASA.

summarizes agency progress in maintaining and improving the geodetic infrastructure since 2010, as well as aspirations for future improvements (Task 1). Chapters 3–7 discuss five categories of science questions (sea-level change, terrestrial water cycle, geological hazards, weather and climate, and ecosystems), the associated measurements that rely on an accurate terrestrial reference frame, and their geodetic needs (Tasks 2 and 3). Detailed connections between the scientific and geodetic needs are presented in Science and Applications Traceability matrixes in Appendix A. Chapter 8 sets priorities for improving the geodetic infrastructure to facilitate answers to the science questions (Task 4) and presents conclusions on all four tasks. The report ends with a list of meeting and workshop participants (see Appendix B), biographical sketches of committee members (see Appendix C), and acronyms and abbreviations used in this report (see Appendix D).

REFERENCES

Adhikari, S., E.R. Ivins, and E. Larour. 2015. ISSM-SESAW v1.0: Mesh-based computation of gravitationally consistent sea level and geodetic signatures caused by cryosphere and climate driven mass change (Data set). https://doi.org/10.5194/gmdd-8-9769-2015.

Altamimi, Z., P. Rebischung, L. Métivier, and X. Collilieux. 2016. ITRF2014: A new release of the International Terrestrial Reference Frame modeling nonlinear station motions. Journal of Geophysical Research: Solid Earth 121(8):6109-6131.

Haines, B.J., Y.E. Bar-Sever, W.I. Bertiger, S.D. Desai, N. Harvey, A.E. Sibois, and J.P. Weiss. 2015. Realizing a terrestrial reference frame using the Global Positioning System. Journal of Geophysical Research: Solid Earth 120(8):5911-5939.

Moreaux, G., F.G. Lemoine, H. Capdeville, S. Kuzin, M. Otten, P. Štěpánek, P. Willis, and P. Ferrage. 2016. The International DORIS Service contribution to the 2014 realization of the International Terrestrial Reference Frame. Advances in Space Research 58(12):2479-2504.

Morel, L., and P. Willis. 2005. Terrestrial reference frame effects on global sea level rise determination from TOPEX/Poseidon altimetric data. Advances in Space Research 36(3):358-368.

NASEM (National Academies of Sciences, Engineering, and Medicine). 2018. Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space. Washington, DC: The National Academies Press.

Nothnagel, A., T. Artz, D. Behrend, and Z. Malkin. 2017. International VLBI service for Geodesy and Astrometry. Journal of Geodesy 91(7):711-721.

NRC (National Research Council). 2010. Precise Geodetic Infrastructure: National Requirements for a Shared Resource. Washington, DC: The National Academies Press.

Pearlman, M.R., J.J. Degnan, and J.M. Bosworth. 2002. The international laser ranging service. Advances in Space Research 30(2):135-143.

Pearlman, M., D. Arnold, M. Davis, F. Barlier, R. Biancale, V. Vasiliev, I. Ciufolini, A. Paolozzi, E.C. Pavlis, K. Sośnica, and M. Bloβfeld. 2019. Laser geodetic satellites: A high-accuracy scientific tool. Journal of Geodesy 93(11):1-14.

Tapley, B.D., G.H. Born, and M.E. Parke. 1982. The SEASAT altimeter data and its accuracy assessment. Journal of Geophysical Research 87(C5):3179-3188.